

Revision Note

Paper: Receiver-Domain Behavioral Probing for Backdoor-Resilient Federated GPS Spoofing Detection in UAV Networks
Authors: Will Jedrzejczak, Cole Walther, Dilpreet Gill, Khalid Hasan
Seven pages to six by selection, not compression. Margins, font sizes and reference formatting are unchanged.

Technical corrections

- **Chai et al. [4] verified against the source.** Their Section IV specifies that each client computes its accuracy on its own local validation set, forms the weight itself, and transmits it to the server, which applies it with no verification. Their Eqs. (18)-(20) are our Eq. (1), λ and Eq. (2). Section II-A now says this and cites their equation numbers. (Their abstract says "standardized validation set," contradicting their own Section IV; we follow Section IV.)
- **Trigger wording.** "In-distribution" removed. The modified feature lies within its own benign marginal range; we add that we do not claim the row is jointly in-distribution.
- **Probe explanation corrected.** The paper no longer implies honest models label every probe row spoofed. Honest recall on a slice is well below one; the mechanism reads a feature-specific deficit relative to the cohort.
- **Adaptive attacker softened** to the evasion objective actually tested, with an explicit note that we do not claim it holds for every defense-aware adversary.
- **Attacker count** is now "no exact attacker count, assuming an honest majority" throughout.
- **Krum terminology separated.** Table I dagggers Krum and Multi-Krum as exclusion from the selected set; all other entries are client flagging. Section II-C defines both.
- **One overhead figure** , 38.6 ms from Table I. The separate 34.0 ms timing run is removed.
- **Weak detector.** Paired lift controls for baseline performance within the evaluated configuration; transfer to a better-converged detector is future work. The causal claim about the federated configuration is withdrawn, since its supporting numbers were cut.

An overclaim sweep of the compiled text (guarantee, prove, eliminate, always, never, cannot, every defense) left only factual statements and two disclaimers.

Cuts

Section	Before	After
Experimental setup	preprocessing commentary	parameters only
Per-client attribution	one subsection	three sentences
Adaptive attacker, root set, parameters, cost	four paragraphs	one paragraph
Discussion and Conclusion	three paragraphs	Conclusion, 178 words

No figure or table removed. Fig. 2 redrawn at one column keeping all five series. Fig. 1 is regenerated: the previous diagram set its labels smaller than the 8 pt caption beneath it and wrote w and L where the equations use ω and f. It is now produced by `build_figures.py` at the same type size as Figs. 2-4, with notation from Eqs. (1)-(7).

Style

One pass over the manuscript removing tutorial framing, including "The message is not that...", "Read from the inside out...", "Why two layers.", "This is the lever the paper is about." and "We are deliberate about scope."

Consistency and reproducibility

Every number in Table I and in the prose is checked against its exported CSV by script. Attack parameters, seeds and defense settings appear once, in Section IV. 15 references, all cited, markers [1]-[15] contiguous. All four figures are written at exactly the width they are placed at, so LaTeX applies no rescaling. `14_conference_results.ipynb` executes with no errors and reproduces Table I and Figs. 2-4 from the CSVs; `build_figures.py` regenerates all four figures. Abstract, contributions, results and conclusion carry the same scoped claims.

Not done: the optional rerun with a better-converged detector. **To check before submission:** volume and page ranges for the three references added this round, [8] Yin, [14] Kang, [15] Kairouz.